TOSHAN GOSWAMI

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Ambitious Computer Engineering student and podium-finishing hacker, placing 2nd at LexHack '26 with a physics-informed wildfire simulator. Builds with hardware and software using IMUs, motor drivers, RF modules in autonomous systems, and training CV/ML pipelines.

EDUCATION

University of Michigan, Ann Arbor, MI **August 2026 -**

Bachelor of Science in Engineering in Computer Engineering

Acton Boxborough Regional High School **Aug 2022 - May 2026**

High School Diploma, GPA 3.92 / 4.0

- Class Leader
- SAT 1530/1600, ACT 34/36
- Honor Roll (all semesters)

HONORS AND AWARDS

- 2nd place of 50 participants, LexHack ‘26 Hackathon, WindMesh Wildfire Spread Simulator
- AP Scholar with Distinction
- Presidential Gold Award for Community Service
- USA Badminton: Gold, Northeast Open Regional Championship

PROJECTS

Airframe AE - Real-Time Acoustic Structural Impact Detection | CSI Hacks 2026 **July 2026**

Built a real-time localization model in Python that processes sensor data to triangulate and visualize structural impacts on a live aircraft diagram.

Arduino RC Car **July 2026**

nRF24L01 wireless communicator, L298N motor driver, MPU6050 IMU; autonomous pathing in development

WindMesh - Wildfire Spread Simulator **June 2026**

Rothermel fire spread model + Computer Vision used to predict wildfire spread in real-world environments

Badminton ML Pipeline **Mar 2026**

YOLOv8 + pose estimation + RandomForest classifier, reached 89% accuracy with hundreds of manually labeled frames

EXPERIENCE

Systems Engineering Intern: SecuriT Locks **August 2026**

Connected sensors to lock software and deployed to pilot; built QR code provisioning system to simplify hardware configuration

Software Engineering Intern: ScoutOut **June 2025 - July 2025**

Built AI model with team to optimize real estate scouting strategy, presented findings to CEO

SKILLS

- **Languages:** Java, Python
- **Tools:** YOLOv8, OpenCV, Arduino
- **Hardware:** IMU sensors, motor drivers, RF modules